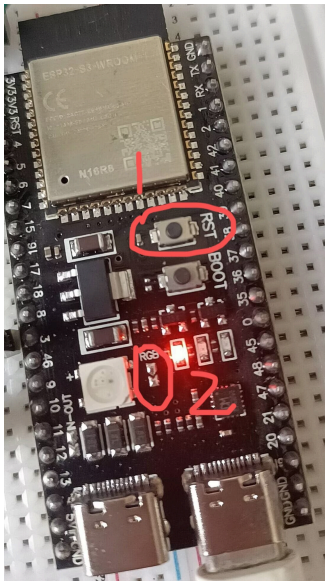
Configure Device Wi-Fi and Add Device and Signal Issues

1. WI-FI Network Configuration

1) Start the device

After downloading/ burning the bin firmware, keep the device powered on, press the RST button on the development board to reset and restart the device (as shown in position 1 in the figure below), and the device will enter the network configuration mode.

Note: If you flash firmware version 0.3.1 or higher, the program will run automatically without the need to manually press the RST button.



2) Distribution Network Status

If the sRGB color light is blue (the white light on the development board) and keeps blinking, it indicates that the device is in the network configuration state.

If the sRGB color light remains unlit, please refer to the detailed instructions in Section 2 of this page.

If the device is not in the network configuration state or needs to be reconfigured, please press and hold the network configuration button on the device (connect pin 1 and GND, for older versions below v0.2.2), while pressing the reset (RST) button on the development board, then release the reset button first, and then release the network configuration button.

3)**Reconfigure the network**

If the device firmware version >= 0.2.2, after failing to reconnect to the original Wi-Fi network after 3 retries, it will re-enter the network configuration mode (you may need to press the RST button to restart the device when switching networks).

If the device firmware version is relatively new, after the device is powered on, directly press and hold the boot button (the button below position 1 in the figure above), and the device will reconfigure the network and no longer connect to the previous WIFI network.

4)Network Configuration Steps

(1) Connect to the Xiaozhi-XXXX hotspot

Use your phone or computer to connect to the device's WiFi network, usually named Xiaozhi-XXXX. Select and connect, wait a few seconds for the WiFi hotspot to connect successfully, and it will automatically enter the selection interface. Do not manually close the interface prematurely, as shown in the following figure:

Note:

If the program does not automatically open a browser to access http://192.168.4.1 , please ensure that the network connected to the wireless local area network (WIFI) is Xiaozhi-XXXXXX and do not switch , use browser address bar to enter the URL http://192.168.4.1 to access the device's network configuration page.

(2) Select and configure WIFI

Please select the user's home or office WIFI network from the 2.4G networks scanned by Xiaozhi Hotspot (no need to manually enter the wifi name SSID), and it will be automatically filled in after the name is selected.

Enter the currently selected Wi-Fi password in the password input box, click the **Connect** button to submit (if it is a password already saved on the phone, it will be submitted automatically).

Note: Opening the Wi-Fi scanning page may be a bit slow. Please wait for about 5 seconds and do not cancel or close the page.

(3) Connect and restart the device

On the network configuration page,**select the WiFi name**(only 2.4G is supported, and the "Maximum Compatibility" option must be enabled for iPhone hotspots), the SSID will be automatically filled in, enter the password below, and then click the **"Connect"** button (as shown in the figure above).

If the Wi-Fi name and password are correct, the device will automatically connect to Wi-Fi and display **Login** success indicator, indicating that the network configuration has been successful, and the device will automatically restart after 3 seconds.

After restarting, the device will automatically connect to the WIFI network configured just now.

If the configuration fails, please check whether the WiFi name and password are correct, or try reconfiguring.

2. How to add a new device to the management backend

After your device is connected to Wi-Fi or 4G, you need to go to the Xiaozhi AI Chatbot - Control Panel to add and verify the device. Only after verification is successful can you use the device for AI voice chat normally. Please follow the steps below:

Please first ensure that the device is connected to the Internet, wake it up via "Hello, Xiaozhi", and wait for the prompt asking to add the device and the 6-digit device verification code (you can repeat the wake-up to listen again).

If you are using a device without offline wake-up, such as the esp32-C3 series development board or product, please use the button to wake up the device for communication, and add the device to the

Xiaozhi AI control panel based on the 6-digit verification code announced by voice or displayed on the screen.

Note: If the user does not hear the voice broadcast and the speaker does not produce sound, users of the breadboard solution should check whether the wiring of the power amplifier and speaker is normal, and users of third-party development boards should check whether the firmware is matched and running properly.

Computer browser access, Xiaozhi AI chatbot - control panel, background: https://xiaozhi.me , if not registered, please register an account first.

Step 1: First, create an agent

Agent Description: Refers to an autonomous entity with artificial intelligence (AI) capabilities, capable of perceiving the environment, processing information, making decisions, and executing actions to achieve specific goals or tasks. It typically possesses characteristics such as interaction ability, perception ability, learning and adaptation ability, autonomy, and goal orientation.

Xiaozhi AI Agent is a combination of physical entities (device terminals and firmware) + virtual entities (backend servers), and is currently mainly used in fields such as AI chatbots, IOT device control, learning and education, and intelligent assistants.

Note: A Xiaozhi AI agent can be connected and used across multiple device terminals. For example, the Xiaozhi AI device you have at home and the one at the office can share the same agent, corresponding to the same AI character, configuration, memory, etc.

Click "+ New Agent" below the navigation on the left side of the page

After confirming the submission, once the new agent is created, it will appear in the agent list, as shown in the following figure:

Step 2: Add a new device

First-time device addition: Please select the "**Add Device**" button on the agent module, click to enter the "**Add New Device**" entry point.

In the input prompt, indicate the 6-digit device ID, then click the "Add Device" button, as shown in the following figure.

Device has been added: Please click "**Device Management**" below the corresponding agent module to enter the device list page, then add a new device via the "**+ Add Device**" button on the right, as shown in the following figure:

After the device is added, it will be automatically activated and displayed on the "Device Management" list page, as shown in the figure above, and then it can be used normally. Currently, the test server is temporarily free of charge.

3. Regarding the WIFI signal issues of some ESP32S3 development boards

Recently, some group members in the QQ group have been replicating the Xiaozhi AI chatbot. After flashing the firmware program and attempting to connect to Wi-Fi, they cannot find the Wi-Fi named Xiaozhi-xxx.

First, confirm that the wifi firmware bin file of the esp32s3 development board has been successfully flashed. When configuring the network after restarting, the white inner circle circular RGB colored lights on the development board, **the network configuration status is a flashing blue light (at a uniform speed)**. For those without soldered RGB lights or those not flashing blue lights, if you are unsure, you can check the log using a serial port tool. To view the serial port log, you can refer to: Operating Instructions for the Independent Serial Port Tool

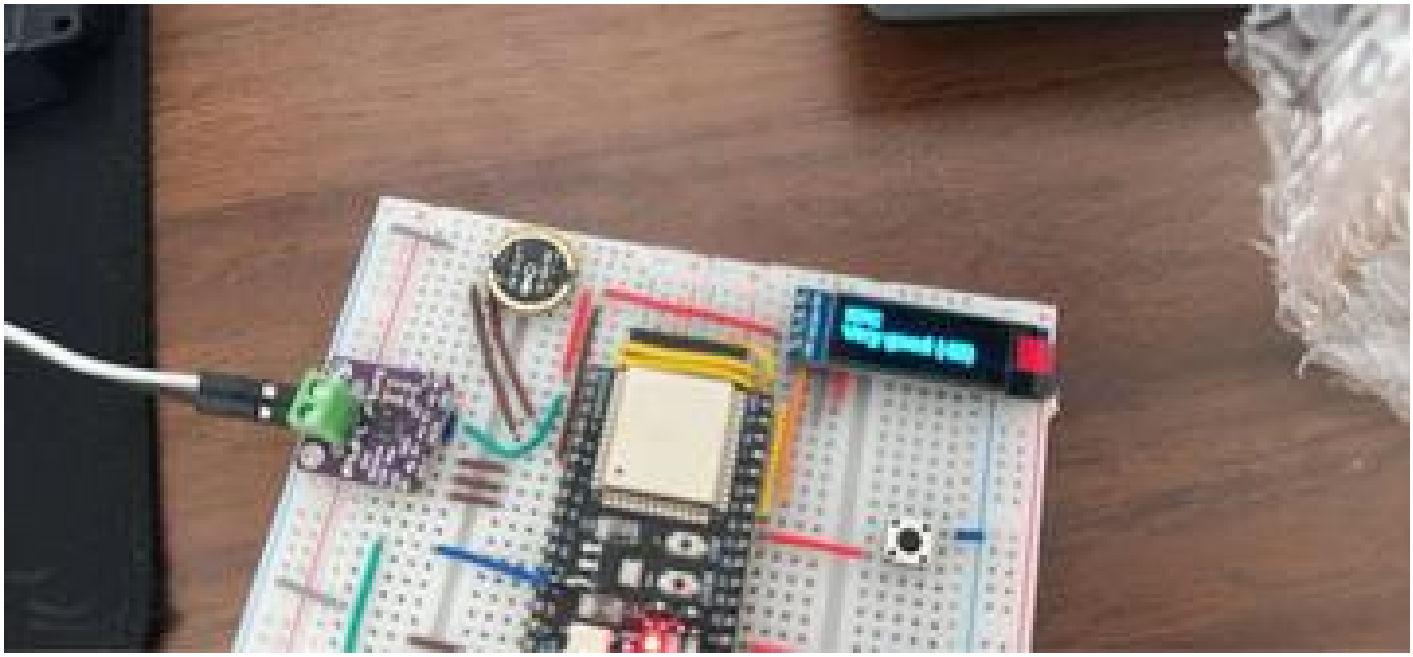
Confirmed that the program burning has no issues, encountered problems with unstable WIFI signals, even resulting in the inability to connect to WIFI. The cause of this problem may be that recently, some product batches from certain merchants have quality issues and were sold without strict inspection.

After everyone's collective wisdom, the following suggestions are presented first as follows :

Recommended Solution:

- **Place the coil:** Use a soft wire to wrap around the antenna position of the ESP32S3 module on the development board about 3 turns, which can enhance the signal reception effect of the antenna. As shown in the position of the yellow wire coil in the figure below.

- **Temporary Solution:** For short-term internet connection needs, please move closer to a location with better WIFI router signal, or temporarily touch the antenna position by hand to connect to WIFI.



Refer to the position of the yellow coil (the picture is from the QQ group, please contact the administrator if you have any questions.)

4. About the sRGB colored lights on the device

The sRGB colored lights on the device (located on the development board, with a white appearance) can display multiple colors, indicating different states of the device:

After powering on, it first flashes blue once, indicating that the device is connecting to WiFi. Then it flashes green once, indicating that the device has connected to WiFi and can now be voice-activated.

After powering on, it first flashes blue once, indicating that the device is connecting to WiFi. Then it remains constantly blue, indicating that the device is performing an OTA firmware update, which usually completes within one minute when the network is good.

After power-on, the blue light keeps flashing: indicating that the device is in the network configuration state.

After voice wake-up, the blue light will turn on, indicating that the device is connecting to the server.

When the green light is on, it indicates that the device is playing voice.

When the red light is on, it indicates that the device is recording.

Regarding the issue of sRGB color lights not lighting up

If the sRGB color light switch is not turned on, the light will not illuminate. Although the device can still be used and it does not affect network configuration, the user experience will be affected, as it will be unable to display the current network configuration status, AI conversation status, etc. In this case, it is recommended to solder it on yourself or find someone to do it (for example, find a store that repairs mobile phones or computers) at the position 2 in the figure below .

*Other issues: It could be a problem with the circuit connection. You can power off, unplug all modules, check the circuit, and directly power on the esp32 s3 development board to see if the blue colored lights flash normally. After ensuring the circuit connection is correct, proceed with the network connection test.

Note: Since this project uses sRGB color lights when using the development board, it is recommended to check the product detail image or consult the seller to see if the sRGB color light switch has been soldered when purchasing the development board. Those who can solder it themselves can ignore this.